# Quiz 5

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An online service receives requests from three types of customers, labeled $A$, $B$ and $C$, at an average rate of $r_A = 0.5$, $r_B = 5$ and $r_C = 2.5$ per minute respectively, independently of other types.

a. What is the probability, $p$ that at least $1$ requests of type $A$ will be received within a minute? 0.3935 ✔

b. What is the probability that the waiting time, $T_A$ between two consecutive requests of type $A$ will be less than $120$ seconds? 0.6321 ✔

c. If the type is ignored, what is the average waiting time (in minutes) between consecutive requests? 0.125 ✔

a. Let $N_A$ = Number of requests of type $A$ within a minute. Then $N_A \sim$ Poisson$(0.5)$ and

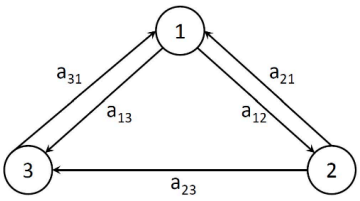
$$\begin{aligned} p &= \mathbb{P}[N_A \geq 1] \\ &= 1 - \mathbb{P}[N_A < 1] \\ &= 1 - \text{ppois}(0, 0.5) = 0.393 \end{aligned}$$

b. By the inevitability of the exponential distribution, $T_A \sim$ Exp$(0.5)$. We are interested in the event $[T_A < 120 \text{ seconds}] = [T_A < 120/60 \text{ minutes}]$. Therefore, the requested probability is:

$$\mathbb{P}[T_A < 120/60] = \text{pexp}(120/60, 0.5) = 0.632$$

c. If the type of request is ignored, the waiting time $T = \min\{T_A, T_B, T_C\}$ follows an exponential distribution with parameter $r_A + r_B + r_C = 8$, therefore $\mathbb{E}[T] = 1/8 = 0.125$ minutes.

Consider the CTMC $X$ represented in the following graph, with transition rates $a_{12} = 9$, $a_{21} = 7$, $a_{13} = 1$, $a_{31} = 8$, $a_{23} = 3$:




a. If the chain is currently at state $1$, what is the probability that the next jump will be to state $3$? 0.1 ✔

b. After jumping to state $3$, how much time is the chain expedted to stay before the next jump? 8 ✖

c. Is the chain irreducible? (Answer $1$ for yes or $0$ for no) 1 ✔

a. The generator of the process is

$$\mathbf{G} = \begin{bmatrix} -10 & 9 & 1 \\ 7 & -10 & 3 \\ 8 & 0 & -8 \end{bmatrix}$$

If the chain is currently at state $1$, the probability that the next jump will be to state $3$ is $-a_{1,3}/a_{1,1} = 0.1$.

be to state 3 is $q_{1,3}/q_{1,1} = 0.12$.

b. The holding time $U_3$ represents the chain stays in state $3$. Since $U_3 \sim$Exp $(8)$, the required expected time is

$$\mathbf{E}[U_3] = \frac{-1}{-8} = 0.125$$

c. The chain is irreducible, since all states can be reached from all others.